StudyForge primer: what is RAG?

Retrieval-augmented generation (RAG) is a way to make a language model answer questions about your own documents instead of only its training data. The idea is simple. First, you break a PDF into chunks and store them in a vector database. When a student asks a question, you retrieve the most relevant chunks and put them in the prompt. The model must answer from those notes and cite the page it used. StudyForge uses RAG so a student can upload lecture slides or a textbook chapter and get answers grounded in that material. Without retrieval, the model might hallucinate a confident-sounding explanation that is not in the notes. A good RAG answer names the page. Example: "A vector database stores embeddings so similar chunks can be found by cosine similarity [p. 2]."

Chunking and embeddings

Chunking means splitting a document into pieces small enough to retrieve. StudyForge keeps chunks on a single page whenever possible so citations stay honest. A typical chunk is a few paragraphs, with a short overlap so a sentence near a boundary is not lost. Embeddings turn each chunk into a list of numbers. Nearby meanings land near each other in that space. StudyForge uses a small sentence-transformers model on CPU (all-MiniLM-L6-v2) so embedding never burns GPU credits. The vector store is ChromaDB, saved on disk under data/chroma. Ingest once, then ask many questions without re-reading the PDF. If you ingest a second file, both sources sit in the same collection and each citation still includes the filename. If a page is empty (for example a slide that is only a diagram), it is skipped.

Citations, quizzes, and explanation modes

Every StudyForge answer should cite pages like [p. 3]. If the notes do not contain the answer, the model should say so instead of guessing. That is the difference between a study copilot and a generic chatbot. Quizzes are generated from the same retrieved notes. A quiz item must be answerable from the PDF. Each question records a source_page so a student can flip back to the lecture slide. Three explanation modes are just different system prompts: - kid: explain like I am 12, with analogies and almost no jargon. - exam: precise vocabulary a grader would expect. - deep: mechanism, edge cases, and nearby topics, still grounded in the notes. This page exists so retrieval tests have a distinct citation. If you ask "How do quizzes stay grounded?" the retriever should prefer page 3.

Why this runs on an AMD Instinct MI300X

The retrieval stack above runs on a laptop CPU. The large language model does not. StudyForge talks to any OpenAI-compatible API. During Week 0 that is a cheap endpoint such as Fireworks. During Week 2 the same code points at a vLLM server on an AMD Instinct MI300X in the AMD Developer Cloud. vLLM on ROCm serves Qwen. A single MI300X has 192 GB of HBM, which easily fits Qwen3-8B or Qwen3-32B with long context. That is the student-budget model. The August 12, 2026 Day-0 announcement was for Qwen3.8-2.4T-A95B, a 2.4 trillion parameter mixture-of-experts model. It needs a multi-GPU node (on the order of 8x MI355X for the MXFP4 recipe), not one droplet and not a $100 credit pack. StudyForge uses the same serving stack ? vLLM + ROCm on Instinct ? with a Qwen that actually fits. Reviewers prefer that honesty over claiming a 2.4T model on a single card. Golden rule: shut the GPU droplet down when you step away. Credits expire 30 days after activation and idle hours still count.